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The weakest floor decides: how soft-story buildings fail, and how a steel frame fixes them

The free guide to the reel on soft-story buildings: what makes a ground floor 'soft', why the sway lands on it, what retrofits are used, and a 10-minute cardboard test you can do at home.



What a soft story is

A soft story is a level that is much less stiff than the one above it. Engineers use a simple test: a level is soft if it is less than 70% as stiff as the floor above it, or less than 80% as stiff as the average of the three floors above.

The classic case is an apartment block with open parking or shop fronts at street level and homes above. The upper floors have many walls. The ground floor has few, because walls would block the cars or the windows.

Why the sway lands on one floor

In an earthquake the ground moves sideways and the building has to follow. Stiff floors move as one block, so they barely bend. The soft floor is the only part that can bend, so almost all of the sideways movement is squeezed into it.

The floors above are heavy, so they keep pushing on the soft floor. Once it leans far enough, it cannot carry them any more and folds. In the 1989 Loma Prieta earthquake, soft-story failures made up nearly half of the homes in the San Francisco Bay Area that became uninhabitable.

Built examples

Northridge Meadows, Los Angeles (1994)	A three-storey apartment block whose ground floor gave way in the Northridge earthquake; 16 people died there, according to news reports of the time.
San Francisco Soft Story Program (2013)	The city required owners of wood-frame buildings of this type to be screened and retrofitted, with work due between 2017 and 2020.
Los Angeles Ordinance 183893 (2015)	Requires owners of older wood-frame buildings with soft, weak or open-front ground floors (most built before 1978) to retrofit them, in phases over several years.

How the fix works

The fix does not make the building stronger everywhere. It makes the weak floor about as stiff as the floors above, so the sway is shared instead of concentrated.

In the garage opening, builders bolt in a steel frame (or add braces or strong plywood walls). The frame is stiff in the sideways direction but still leaves room for the car. In the reel, you see the sway spread over the three floors once the frame is in.

Try it yourself: a cardboard soft story

1. Take a tall cardboard box (a cereal box works) and stand it on a baking tray.
2. Cut a large opening in the bottom third of one long side (your garage). Leave the upper two thirds untouched.
3. Put a couple of books on top so the weight is realistic.
4. Slide the tray sideways back and forth, a little faster each time, and watch which part bends first. It will be the cut-open bottom.
5. Now tape two cardboard diagonals inside the opening (an X) and repeat. The box takes a much bigger shake before it leans. That X is your steel frame.

The rule to remember

A structure is only as safe as its weakest floor, so check the weak spot before you add strength elsewhere.

Default answer	Make the whole building stronger.
Better answer	Find the one level that moves most and stiffen only that one.
Check	Ask: where would the sway land if the ground moved sideways today?

Go further

Soft story building (Wikipedia)
https://en.wikipedia.org/wiki/Soft_story_building

1994 Northridge earthquake (Wikipedia)
https://en.wikipedia.org/wiki/1994_Northridge_earthquake

FEMA: earthquake risk management
https://www.fema.gov/emergency-managers/risk-management/earthquake

Credits

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Facts: Wikipedia (soft story building), news reports of the Northridge Meadows collapse, City of San Francisco and City of Los Angeles ordinances.

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